

Vijaya Lakshmi D S

B.E. Electronics & Communication Engineering | Software Developer | AI, Cloud, Computer Vision & IoT

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PROFILE

B.E. ECE student focused on software development, with hands-on work across Python, React, FastAPI, cloud-backed data systems, AI/ML pipelines, and IoT-integrated products. Builds practical systems for healthcare access, computer vision, safety, automation, and local AI workflows.

TECHNICAL SKILLS

Software Development: Python, JavaScript, TypeScript, React, FastAPI, REST APIs, HTML/CSS, Git/GitHub, Docker, Linux

Cloud & Data: Google Cloud, AWS Bedrock/RDS, Azure, Firebase Auth/Firestore, PostgreSQL, MySQL, MongoDB, Redis/Valkey

AI/ML: TensorFlow/Keras, OpenCV, NumPy, Pandas, scikit-learn, Matplotlib, LangChain, LangGraph, Ollama/local LLM workflows

Embedded & IoT: C, Arduino IDE, microcontrollers, sensor integration, WebRTC-connected device workflows

EXPERIENCE

Computer Vision Internship Project | SSN College of Engineering

Jun 2026

- Built a person-wise HAR pipeline as part of an internship project, combining YOLOv8n detection, reproducible preprocessing, and transfer-learning classifiers.
- Prepared 6,097 retained source images into 12,262 person crops; compared EfficientNetV2B3, VGG16, and VGG19, with EfficientNetV2B3 reaching 84.05% accuracy and 82.98% macro F1.

Embedded Systems Intern | CodeBind Technologies, Chennai

Jun 2025

- Worked on microcontroller programming and hardware-software integration using embedded C and Arduino workflows.

UBA Intern | Sri Sairam Techno Incubator Foundation

Apr 2025

- Contributed to technical surveys and rural-development initiatives, connecting engineering work with community use cases.

Web Development Intern | OctaNet Services Pvt. Ltd.

Jun 2024

- Built responsive web pages using HTML, CSS, and JavaScript, with attention to layout, usability, and browser behavior.

SELECTED SOFTWARE PROJECTS

AI-Assisted Telemedicine Kiosk | Python, Firebase, WebRTC, IoT sensors, generative AI, healthcare workflows

- IEEE EPICS grant recipient, USD 2,910; building a rural healthcare kiosk/software flow for ABHA-based identification, vital capture, symptom mapping, WebRTC consultation, and QR-linked reports.
- Designed around Firebase-backed records, doctor verification, multilingual access, print-ready prescriptions, AI-assisted medication suggestions, and sensor input from MLX90614, MAX30102, ultrasonic, and smart-scale modules.

Pratiksha WhatsApp AI Companion | TypeScript, Docker, Postgres, pgvector, Ollama

- Contributed to a local-first AI companion concept using WhatsApp workflows, local LLMs, vector memory, resource matching, and dashboard controls.

Collaborative Builds | Kotlin, Jetpack Compose, Firebase, Python, Google Calendar API, AI planning

- Contributed to Wordsworth, a Firebase-backed Android vocabulary game; Pournami Calendar Automation, a Python worker for Google Calendar sync; and SatarkAI planning for Gmail scam detection and local LLM scoring.

RESEARCH, HARDWARE, AND SUPPORTING WORK

- Controlled maxillofacial administration of Levodopa for Parkinson's disease: contributed Python/Matplotlib data visualization for medical data; research published in Cureus.
- Built sensor-driven elderly care and automatic radar prototypes using IoT monitoring, Arduino IDE, and ultrasonic sensing.

EDUCATION

Sri Sai Ram Engineering College, Chennai | B.E. Electronics and Communication Engineering

2023-2027

Academic focus: software development, electronics, embedded systems, communications, computer vision, and AI-enabled systems.

ACHIEVEMENTS AND CERTIFICATIONS

- IEEE EPICS grant recipient, USD 2,910, for the AI-Assisted Telemedicine Kiosk.
- AI-Assisted Telemedicine Kiosk paper selected for presentation at AICCoNS 2025.
- Cisco Data Science certification; Microsoft Generative AI certifications; Udemy coursework in web development, AI, and Python.